

Ten Essential Concepts in Probability

A rigorous study guide for competition preparation and review

Abstract

Probability measures how likely an event is to occur. For any event A , its probability satisfies $0 \leq P(A) \leq 1$, where 0 means impossible and 1 means certain. Probability questions begin by identifying the random experiment, the sample space S of elementary outcomes, and the event $A \subseteq S$ of interest. These notes develop ten essential ideas: estimating probability from experiments, computing probabilities from finite models, using sets and conditional probability, counting with permutations and combinations, connecting events to random variables, and recognizing continuous, binomial, and geometric probability models. The emphasis is on choosing a valid model, checking its assumptions, calculating exactly when possible, and interpreting the answer.

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1 Experimental Probability

A *random experiment* is a process whose outcome is uncertain before it is performed. One performance of the experiment is a *trial*. An *outcome* is a possible result; an *elementary outcome* is one indivisible result in the chosen model. The *sample space* S is the set of elementary outcomes under consideration, and an *event* is a subset of S .

Vocabulary example

For one roll of a fair die, the experiment is rolling the die once, one trial is one roll, the elementary outcomes are 1, 2, 3, 4, 5, 6,

$$S = \{1, 2, 3, 4, 5, 6\}, \quad A = \{1, 3, 5\}$$

is the event “an odd result.”

Experimental probability is a probability estimated from observed data. If event A occurs in $n(A)$ of N trials, then the experimental probability, or relative frequency, is

$$\hat{P}(A) = \frac{n(A)}{N}, \quad N > 0.$$

This formula is valid after the trials have actually been performed.

1.1 Worked example: flipping a coin

Suppose a coin is flipped 10 times and tails occurs 3 times. Then

$$\hat{P}(\text{tails}) = \frac{3}{10} = 0.3 = 30\%.$$

This means that tails occurred in 30% of the flips in this particular experiment. It does *not* mean that the coin’s theoretical probability of tails is 0.3.

1.2 Experimental versus theoretical probability

For a fair coin, the theoretical probability of tails is $1/2 = 0.5 = 50\%$. Experimental probability is a sample proportion, so repeated experiments with the same number of trials can produce different relative frequencies. For independent repeated trials with the same event probability $p = P(A)$, the relative frequency converges toward p in the long run. Individual experiments can still fluctuate, and adding more trials does not guarantee monotonic improvement after every trial; larger sample sizes generally reduce typical fluctuation.

Table 1: Experimental and theoretical probability.

Type	Based on	Main caution
Experimental probability	Observed outcomes from trials	May vary from one experiment to another
Theoretical probability	A mathematical model of the sample space	Requires correct assumptions about the model

Simulation uses artificial repetitions of a model to estimate a probability that is difficult to calculate directly. For example, to estimate the probability of getting at least one six in five die

rolls, one could repeatedly simulate five fair rolls, record whether at least one six appeared, and compute the relative frequency. Simulation estimates the model probability; it does not replace the need to check whether the model is appropriate.

Key idea

Experimental probability answers, “What happened in these trials?” Theoretical probability answers, “What should happen according to the model?”

2 Theoretical Probability

Theoretical probability uses a mathematical description of the experiment to compute the probability of an event. In a finite model, probabilities obey

$$P(A) \geq 0, \quad P(S) = 1,$$

and if $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

Consequences include

$$P(\emptyset) = 0, \quad P(A^c) = 1 - P(A), \quad A \subseteq B \implies P(A) \leq P(B).$$

These are finite-level probability rules; no advanced set theory is needed here.

For a finite sample space S whose elementary outcomes are equally likely, the probability of event $A \subseteq S$ is

$$P(A) = \frac{|A|}{|S|}.$$

Here $|A|$ is the number of favorable outcomes and $|S|$ is the total number of possible outcomes. This favorable-outcomes formula is valid only when all elementary outcomes in S are equally likely.

2.1 Worked examples

In these examples, assume fair dice, a well-shuffled standard deck, and random selection.

1. Rolling a four on a fair six-sided die:

$$P(\text{four}) = \frac{1}{6}.$$

2. Rolling an odd number:

$$P(\text{odd}) = \frac{3}{6} = \frac{1}{2} = 0.5 = 50\%.$$

3. Drawing the seven of hearts from a standard 52-card deck:

$$P(\text{seven of hearts}) = \frac{1}{52}.$$

4. Drawing a king:

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13}.$$

5. Drawing a red marble from a bag with 3 red, 2 blue, and 5 green marbles. The elementary outcomes are the individual marbles, not merely the three color names, so

$$P(\text{red}) = \frac{3}{3 + 2 + 5} = \frac{3}{10} = 0.3.$$

2.2 Complements and weighted finite models

The *complement* A^c is the event that A does not occur. For the marble bag above,

$$P(\text{not blue}) = 1 - P(\text{blue}) = 1 - \frac{2}{10} = \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%.$$

The notation A' is also sometimes used for a complement, but A^c is more standard in many probability texts.

If elementary outcomes are not equally likely, add their probabilities instead of counting them. For a finite sample space,

$$P(A) = \sum_{\omega \in A} P(\{\omega\}),$$

where each elementary probability is nonnegative and all elementary probabilities sum to 1.

Table 2: Constructed weighted finite sample space.

Outcome	a	b	c
Probability	0.2	0.5	0.3

In this constructed model, the event $\{a, c\}$ has probability $0.2 + 0.3 = 0.5$, not $2/3$. Counting outcomes alone fails because the elementary outcomes are not equally likely.

Common error

Do not use $P(A) = |A|/|S|$ unless the elementary outcomes are equally likely. A biased coin, a nonuniform spinner, or a weighted table requires probability weights.

3 Probability Using Sets

Probability using sets describes events as subsets of a sample space and uses set operations to combine them. The main notation is as follows.

- S : the sample space, or set of all possible outcomes.
- Event: any subset of S .
- $\emptyset$: the empty event, which contains no outcomes.
- A^c : the complement of A , meaning outcomes not in A .
- $A \cap B$: the intersection, meaning outcomes in both A and B .
- $A \cup B$: the union, meaning outcomes in A or B or both.
- Mutually exclusive events: events with no overlap, so $A \cap B = \emptyset$.

The word *and* usually signals an intersection. The word *or* in probability is inclusive: A or B means A , B , or both.

De Morgan's laws describe complements of combined events:

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

A *partition* of S is a collection of mutually exclusive events whose union is all of S . Partitions are useful because they let us split a probability into disjoint cases.

Table 3: Language translated into event notation.

Phrase	Event notation
A and B	$A \cap B$
A or B ; at least one of A, B	$A \cup B$
Only A	$A \cap B^c$
Neither A nor B ; none of A, B	$(A \cup B)^c = A^c \cap B^c$
Exactly one of A, B	$(A \cap B^c) \cup (A^c \cap B)$

3.1 The addition rule

For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The subtraction removes the overlap, which was counted once in $P(A)$ and again in $P(B)$. If A and B are mutually exclusive, then $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$.

Table 4: Venn-style regions for two events.

Region	Description
$A \cap B$	Outcomes in both events
$A \cap B^c$	Outcomes in A only
$A^c \cap B$	Outcomes in B only
$(A \cup B)^c$	Outcomes in neither event

3.2 Worked example: a sports survey

A survey has 100 respondents. The verified information is: 8 respondents like both hockey and basketball; 48 like basketball; 42 like soccer; and 15 like both basketball and soccer. Let B be “likes basketball” and C be “likes soccer.” The probability that a randomly selected respondent likes basketball or soccer is

$$\begin{aligned} P(B \cup C) &= \frac{48 + 42 - 15}{100} \\ &= \frac{75}{100} = \frac{3}{4} = 0.75. \end{aligned}$$

Thus 75% of respondents are known to be in the basketball-or-soccer union. The value $8/100 = 0.08$ is the probability of liking hockey and basketball. No unreported regions of the original Venn diagram are needed for these conclusions.

Key idea

When using a Venn diagram, add each region exactly once. For a union, subtract the intersection if the totals for two sets have already counted it twice.

4 Conditional Probability

Conditional probability is the probability of one event under the condition that another event has occurred. The probability of B given A is

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

For a finite equally counted sample, this can be written as

$$P(B | A) = \frac{|A \cap B|}{|A|}, \quad |A| > 0.$$

Conditioning restricts the relevant sample space from all of S to the event A .

4.1 Worked example: school preference

The following complete table is a constructed classroom example.

Table 5: Constructed school-preference table.

	Likes school L	Does not like school L^c	Total
Female F	10	4	14
Not female F^c	8	8	16
Total	18	12	30

Then

$$P(L | F) = \frac{|L \cap F|}{|F|} = \frac{10}{14} = \frac{5}{7} \approx 0.7143,$$

while

$$P(F | L) = \frac{|F \cap L|}{|L|} = \frac{10}{18} = \frac{5}{9} \approx 0.5556.$$

The reversed conditionals are different because their denominators are different. These are associations within the data, not causal claims.

4.2 Total probability and Bayes' theorem

If $B_1, \dots, B_m$ form a partition of S and each relevant denominator is positive, then

$$P(A) = \sum_{i=1}^m P(B_i) P(A | B_i).$$

Bayes' theorem reverses a conditional probability:

$$P(B_j | A) = \frac{P(B_j) P(A | B_j)}{\sum_{i=1}^m P(B_i) P(A | B_i)}, \quad P(A) > 0.$$

Base-rate example. A screening test is applied in a population where 2% have a condition. The test is positive for 90% of people with the condition and for 5% of people without it. Let D be “has the condition” and $+$ be “positive test.” Then

$$\begin{aligned} P(+) &= P(D) P(+ | D) + P(D^c) P(+ | D^c) \\ &= (0.02)(0.90) + (0.98)(0.05) = 0.067. \end{aligned}$$

Thus

$$P(D | +) = \frac{(0.02)(0.90)}{0.067} = \frac{18}{67} \approx 0.2687.$$

The base rate $P(D) = 0.02$ matters; $P(+ | D)$ and $P(D | +)$ answer different questions.

Common error

In $P(B | A)$, the denominator is the size or probability of the condition A , not the entire original sample space.

5 The Multiplication Law

The *multiplication law* computes the probability that events occur together. Whenever the conditional probabilities are defined,

$$P(A \cap B) = P(A) P(B | A) = P(B) P(A | B).$$

Events A and B are *independent* if and only if

$$P(A \cap B) = P(A) P(B).$$

When $P(A) > 0$ and $P(B) > 0$, this is equivalent to

$$P(B | A) = P(B), \quad P(A | B) = P(A).$$

For sequential events,

$$P(A_1 \cap \cdots \cap A_m) = P(A_1) P(A_2 | A_1) \cdots P(A_m | A_1 \cap \cdots \cap A_{m-1}).$$

Table 6: Independent, dependent, and mutually exclusive events.

Relationship	Meaning	Important distinction
Independent	Product rule $P(A \cap B) = P(A) P(B)$ holds	Can overlap
Dependent	The product rule with unconditional probabilities does not hold	Often occurs in sampling without replacement
Mutually exclusive	Cannot occur together	If both have positive probability, they are not independent

5.1 Worked examples

A die and a coin. Rolling a fair die and flipping a fair coin are independent. Therefore,

$$P(\text{three and tails}) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}.$$

Two kings without replacement. Drawing two kings consecutively from a standard deck without replacement is dependent. The first draw has 4 kings among 52 cards. If the first card is a king, then 3 kings remain among 51 cards, so

$$P(\text{two kings}) = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}.$$

The numerator and denominator change because the first card is not put back.

5.2 Replacement and branch tables

A branch table is a text version of a probability tree: multiply along a path and add the paths that satisfy the event. Suppose a bag contains 4 red and 6 blue marbles.

Table 7: Two draws with replacement.

First draw	Second draw	Path probability	Interpretation
<i>R</i>	<i>R</i>	$\frac{4}{10} \cdot \frac{4}{10} = \frac{4}{25}$	probabilities unchanged
<i>R</i>	<i>B</i>	$\frac{4}{10} \cdot \frac{6}{10} = \frac{6}{25}$	probabilities unchanged
<i>B</i>	<i>R</i>	$\frac{6}{10} \cdot \frac{4}{10} = \frac{6}{25}$	probabilities unchanged
<i>B</i>	<i>B</i>	$\frac{6}{10} \cdot \frac{6}{10} = \frac{9}{25}$	probabilities unchanged

With replacement, $P(\text{one red and one blue}) = 6/25 + 6/25 = 12/25$.

Table 8: Two draws without replacement.

First draw	Second draw	Path probability	Interpretation
<i>R</i>	<i>R</i>	$\frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15}$	probabilities change
<i>R</i>	<i>B</i>	$\frac{4}{10} \cdot \frac{6}{9} = \frac{4}{15}$	probabilities change
<i>B</i>	<i>R</i>	$\frac{6}{10} \cdot \frac{4}{9} = \frac{4}{15}$	probabilities change
<i>B</i>	<i>B</i>	$\frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$	probabilities change

Without replacement, $P(\text{one red and one blue}) = 4/15 + 4/15 = 8/15$.

Key idea

Use multiplication for “and” along a path, but decide first whether later probabilities must be conditional. Add disjoint successful paths.

6 Permutations

The *fundamental counting principle* says that if a process has m_1 choices at the first stage, m_2 choices at the second, and so on, then the total number of possible outcomes is

$$m_1 m_2 \cdots m_r.$$

For example, a lunch with 3 sandwich choices, 2 drink choices, and 4 side choices can be selected in $3 \cdot 2 \cdot 4 = 24$ ways.

A *permutation* is an ordered arrangement of objects. For a nonnegative integer n , factorial notation means

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

For $0 \leq r \leq n$, the number of ordered arrangements of r distinct objects chosen from n distinct objects is

$${}_nP_r = P(n, r) = \frac{n!}{(n-r)!}.$$

This formula applies when order matters and repetition is not allowed.

If order matters and repetition is allowed, then each of the r positions has n choices, so there are

$$n^r$$

ordered selections.

6.1 Worked examples

The six arrangements of A, B, C are

$$ABC, ACB, BAC, BCA, CAB, CBA.$$

For first-, second-, and third-place finishes among 10 runners,

$${}_{10}P_3 = \frac{10!}{7!} = 10 \cdot 9 \cdot 8 = 720.$$

For a lock code of length 3 using numbers 0 through 59:

- if repetition is allowed, there are $60^3 = 216,000$ possible codes;
- if the three numbers must be distinct, there are

$${}_{60}P_3 = 60 \cdot 59 \cdot 58 = 205,320$$

possible codes.

If only one distinct-number code is correct, then

$$P(\text{correct first guess}) = \frac{1}{205,320}.$$

Check: $60 \cdot 59 \cdot 58 = 205,320$.

Key idea

A permutation question asks, “How many ordered lists can be made?” If swapping two selected objects creates a different outcome, order matters.

7 Combinations

A *combination* is a selection in which order does not matter. For $0 \leq r \leq n$, the number of ways to choose r objects from n distinct objects is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The extra factor $r!$ in the denominator removes the different orders of the same selected group. Equivalently,

$${}_nP_r = r! \binom{n}{r},$$

because one may first choose the r objects and then arrange the selected objects in $r!$ orders.

7.1 Worked examples

The number of groups of three from five people is

$$\binom{5}{3} = \frac{5!}{3!2!} = 10.$$

Now suppose a team of 6 is chosen from 7 children and 11 adults. The sample space contains

$$\binom{18}{6} = 18,564$$

teams. To form a favorable team with exactly one child, choose 1 of the 7 children and 5 of the 11 adults:

$$\binom{7}{1} \binom{11}{5} = 7 \cdot 462 = 3,234.$$

Therefore

$$P(\text{exactly one child}) = \frac{\binom{7}{1} \binom{11}{5}}{\binom{18}{6}} = \frac{3234}{18564} = \frac{77}{442} \approx 0.1742.$$

7.2 Complementary counting

Sometimes it is easier to count what does *not* happen:

$$P(\text{at least one}) = 1 - P(\text{none}).$$

For example, the probability of at least one six in two fair die rolls is

$$1 - P(\text{no sixes}) = 1 - \left(\frac{5}{6}\right)^2 = 1 - \frac{25}{36} = \frac{11}{36} \approx 0.3056.$$

Table 9: Permutations and combinations.

Counting method	Use when	Formula
Ordered with repetition	Order matters, repetition allowed	n^r
Permutation	Order matters, no repetition	$nPr = \frac{n!}{(n-r)!}$
Combination	Order does not matter	$\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Connection to binomial probabilities

In a binomial probability, $\binom{n}{k}$ counts the possible locations of the k successes among the n trials.

8 Continuous Probability Distributions

Before named distributions, connect events to numerical values. A *random variable* X assigns a number to each outcome of an experiment. Its *support* is the set of values it can take. A discrete random variable has possible values that can be listed or counted; a continuous random variable has possible values occupying intervals.

For three fair coin flips, let X be the number of heads. Then the support is $\{0, 1, 2, 3\}$; for example, $X(\text{HTH}) = 2$. A *probability mass function* (PMF) is

$$p_X(k) = P(X = k), \quad p_X(k) \geq 0, \quad \sum_k p_X(k) = 1.$$

For two fair coin flips, the PMF of $X = \text{number of heads}$ is: The probabilities are nonnegative and

Table 10: PMF for heads in two fair flips.

k	0	1	2
$P(X = k)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

sum to $1/4 + 1/2 + 1/4 = 1$.

The *cumulative distribution function* (CDF) of any random variable is

$$F(x) = P(X \leq x).$$

Thus $P(X > x) = 1 - F(x)$. For integer-valued discrete variables, endpoints matter: for example, $P(X \geq k) = 1 - F(k - 1)$.

For a discrete random variable,

$$E(X) = \sum_x x P(X = x)$$

is its expected value, interpreted as a long-run average. It need not be a possible individual observation. Variance and standard deviation measure spread:

$$\text{Var}(X) = E[(X - E(X))^2], \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

A *continuous random variable* can take any value in an interval of real numbers. A *probability density function* (PDF) $f(x)$ assigns density, not point probability, and satisfies

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

The integral represents area under the density curve; no integration techniques are needed in these notes. For a continuous random variable X with density f ,

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Thus probabilities are areas under the curve, not curve heights. Also, $P(X = x) = 0$ for an individual value, so endpoints do not affect interval probabilities: $P(a \leq X \leq b) = P(a < X < b)$.

8.1 Uniform distribution

A uniform distribution on an interval gives equal density to equal lengths. We write

$$X \sim U(a, b), \quad f(x) = \frac{1}{b - a} \quad \text{for } a \leq x \leq b.$$

If $X \sim U(0, 10)$, then

$$P(2 < X < 7) = \frac{7 - 2}{10 - 0} = \frac{1}{2} = 0.5.$$

The shaded region in a graph would be the interval from 2 to 7 under a flat rectangle of height $1/10$.

8.2 The normal distribution

A normal model is appropriate only when context or assumptions support an approximately symmetric, bell-shaped distribution; not every continuous variable or every bell-shaped curve is normal. We write $X \sim N(\mu, \sigma^2)$ when X is normal with mean μ , variance σ^2 , and standard deviation $\sigma > 0$. Approximately 68%, 95%, and 99.7% of the area lie within 1, 2, and 3 standard deviations of the mean, respectively, under a normal model.

The standard normal distribution is

$$Z \sim N(0, 1),$$

with mean 0, variance 1, and standard deviation 1. Its CDF is

$$\Phi(z) = P(Z \leq z).$$

To standardize a normal value, use

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Useful identities are

$$P(Z > z) = 1 - \Phi(z), \quad P(a < Z < b) = \Phi(b) - \Phi(a), \quad \Phi(-z) = 1 - \Phi(z).$$

8.3 Worked examples: normal-model calculations

Suppose

$$X \sim N(68, 10^2), \quad \text{and find } P(X > 76).$$

The standardized value is

$$z = \frac{76 - 68}{10} = 0.8.$$

Using a standard normal table or a normal CDF,

$$\Phi(0.8) \approx 0.7881.$$

Therefore

$$P(X > 76) = 1 - \Phi(0.8) \approx 1 - 0.7881 = 0.2119.$$

Under the stated normal-model assumption, the probability is approximately 0.2119, or 21.19%.

For a between-values calculation, suppose $X \sim N(100, 15^2)$. Then

$$P(85 < X < 115) = P(-1 < Z < 1) = \Phi(1) - \Phi(-1) \approx 0.8413 - 0.1587 = 0.6826.$$

For percentiles, if $\Phi(z_q) = q$, then

$$x_q = \mu + \sigma z_q.$$

For example, the 90th percentile of $N(100, 15^2)$ is approximately $100 + 15(1.2816) = 119.224$, because $\Phi(1.2816) \approx 0.9000$.

Common error

A density height is not the probability of an exact value, and a z -score is not a probability. Probabilities come from CDF values or areas over intervals.

Table 11: Discrete and continuous distributions.

Type	Probability tool	Endpoint behavior
Discrete	Probability mass function	Individual values can have positive probability
Continuous	Probability density function and area	Individual values have probability 0

9 Binomial Probability Distributions

A *binomial probability distribution* models the number of successes in a fixed number of Bernoulli trials. The conditions are:

1. a fixed number n of trials;
2. two categorized outcomes per trial, called success and failure;
3. independent trials;
4. a constant success probability p on every trial.

If X is the number of successes, then

$$X \sim \text{Bin}(n, p)$$

and the PMF is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

A particular sequence with k successes and $n - k$ failures has probability $p^k (1 - p)^{n-k}$. The factor $\binom{n}{k}$ counts the possible locations of the k successes among the n trials. The expected value, variance, and standard deviation are

$$E(X) = np, \quad \text{Var}(X) = np(1 - p), \quad \text{SD}(X) = \sqrt{np(1 - p)}.$$

For cumulative probabilities,

$$P(X \leq r) = \sum_{k=0}^r P(X = k),$$

and complements often simplify calculations, such as

$$P(X \geq 1) = 1 - P(X = 0).$$

Binomial checklist

Before using a binomial model, ask: fixed n ? two categorized outcomes? independent trials? constant p ? Sampling without replacement generally changes p and creates dependence, so it is not exactly binomial; a large population may justify an approximation, but that is a modeling judgment.

9.1 Worked examples

Rolling threes. Roll a fair six-sided die four times and define success as rolling a three. Then

$$X \sim \text{Bin}\left(4, \frac{1}{6}\right).$$

Table 12: Number of threes in four rolls.

k	Exact probability $P(X = k)$	Decimal approximation
0	$\left(\frac{5}{6}\right)^4 = \frac{625}{1296}$	0.4823
1	$\binom{4}{1} \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^3 = \frac{500}{1296}$	0.3858
2	$\binom{4}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2 = \frac{150}{1296}$	0.1157
3	$\binom{4}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right) = \frac{20}{1296}$	0.0154
4	$\left(\frac{1}{6}\right)^4 = \frac{1}{1296}$	0.0008

The complete probability distribution is: The exact probabilities sum to

$$\frac{625 + 500 + 150 + 20 + 1}{1296} = \frac{1296}{1296} = 1.$$

The expected number of threes is

$$E(X) = 4 \left(\frac{1}{6}\right) = \frac{2}{3} \approx 0.667.$$

The variance is

$$\text{Var}(X) = 4 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right) = \frac{5}{9}, \quad \text{SD}(X) = \frac{\sqrt{5}}{3} \approx 0.7454.$$

The expected value is a long-run average over many repetitions of the four-roll experiment; it need not be a possible observed count in one experiment.

At least one six. If a fair die is rolled 5 times and X is the number of sixes, then $X \sim \text{Bin}(5, 1/6)$ and

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{5}{6}\right)^5 = 1 - \frac{3125}{7776} = \frac{4651}{7776} \approx 0.5981.$$

A failed binomial checklist. Drawing 5 cards from a deck without replacement and counting aces is not exactly binomial: the number of trials is fixed and there are two categorized outcomes, but the draws are dependent and the success probability changes after each draw.

10 Geometric Probability Distributions

A *geometric probability distribution* models the trial number on which the first success occurs. The conditions are:

1. independent repeated Bernoulli trials;
2. two categorized outcomes per trial;
3. constant success probability p ;
4. trials continue until the first success.

If X is the trial number of the first success, then

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

The expected waiting time and variance are

$$E(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}.$$

Tail and cumulative probabilities are often easiest by complements:

$$P(X > k) = (1 - p)^k, \quad P(X \leq k) = 1 - (1 - p)^k.$$

Here $P(X > k)$ means no success occurs in the first k trials.

10.1 Worked example: rolling doubles

Roll two fair six-sided dice repeatedly until doubles appear. There are 36 equally likely ordered outcomes and 6 doubles, so

$$p = P(\text{doubles}) = \frac{6}{36} = \frac{1}{6}.$$

The probability that the first doubles occur on trial 4 is

$$P(X = 4) = \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right) = \frac{125}{1296} \approx 0.09645.$$

For fewer than three trials,

$$\begin{aligned} P(X < 3) &= P(X = 1) + P(X = 2) \\ &= \frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} = \frac{6}{36} + \frac{5}{36} = \frac{11}{36} \approx 0.3056. \end{aligned}$$

The probability that doubles do not occur in the first five trials is

$$P(X > 5) = \left(\frac{5}{6}\right)^5 = \frac{3125}{7776} \approx 0.4019.$$

The expected waiting time is

$$E(X) = \frac{1}{1/6} = 6,$$

and the variance is

$$\text{Var}(X) = \frac{1 - 1/6}{(1/6)^2} = 30.$$

On average, it takes six rolls of the pair of dice to get doubles.

10.2 Binomial or geometric?

For fair coin flips, “exactly two heads in five flips” is binomial:

$$\binom{5}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32} = \frac{5}{16}.$$

The question fixes five trials and counts successes. By contrast, “the first head occurs on flip five” is geometric:

$$\left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) = \frac{1}{32}.$$

The process continues until the first success.

Situations that are not geometric include drawing from a finite deck without replacement, any process with changing success probability, and an experiment that stops after a fixed number of trials regardless of whether success occurs.

Memoryless property

For a geometric random variable, the chance of waiting more than t additional trials does not depend on already having waited s trials:

$$P(X > s + t \mid X > s) = P(X > t).$$

This property relies on independent trials with constant p .

Table 13: Binomial and geometric models.

Model	Random variable	Experiment stops when
Binomial	Number of successes in n trials	The fixed n trials are completed
Geometric	Trial number of first success	The first success occurs

11 Final Synthesis

The progression of ideas is

experiment $\longrightarrow$ sample space and events $\longrightarrow$ counting and conditioning $\longrightarrow$ random variable $\longrightarrow$ distribution.

Experimental probability estimates likelihood from data, while theoretical probability uses a model. Set notation clarifies intersections, inclusive unions, complements, and partitions. Conditional probability changes the sample space, and the multiplication law turns sequential “and” questions into products, using conditional probabilities when events are dependent.

Permutations and combinations support probability by counting sample spaces and event spaces. Use permutations when order matters and combinations when only the selected group matters. Random variables turn outcomes into numbers. Continuous models assign probability through areas under density curves. The binomial and geometric distributions are discrete PMFs for repeated Bernoulli trials, but they answer different questions: “How many successes in a fixed number of trials?” versus “How long until the first success?”

12 Formula Reference

13 Mixed Review Exercises

1. A coin is flipped 20 times and heads appears 11 times. Find the experimental probability of heads.
2. A card is drawn from a standard 52-card deck. Find the probability that it is not a king.
3. In a class of 30 students, 18 play soccer, 12 play basketball, and 7 play both. Let C be “plays soccer” and B be “plays basketball.” What is $P(C \cup B)$?
4. A bag contains 4 red and 6 blue marbles. Two marbles are drawn without replacement. Find the probability that both are red.
5. How many ways can gold, silver, and bronze medals be awarded among 12 competitors if no competitor can receive more than one medal?
6. How many committees of 4 can be formed from 9 people?

Table 14: Model-selection checklist.

Question to ask	Likely tool
Are elementary outcomes equally likely?	Use favorable outcomes over total outcomes.
Are elementary outcomes weighted unequally?	Add the weights of favorable outcomes.
Is there a condition or restricted sample space?	Use conditional probability.
Are events sequentially dependent?	Use conditional multiplication or a branch table.
Does order matter? Is repetition allowed?	Choose among n^r , permutations, and combinations.
Is the random variable discrete or continuous?	Use a PMF/CDF or a PDF/area model.
Is there a fixed number of Bernoulli trials?	Consider a binomial model after checking assumptions.
Does the process continue until the first success?	Consider a geometric model after checking assumptions.
Are the assumptions actually satisfied?	Revise or reject the model if assumptions fail.

7. Suppose $X \sim N(50, 8^2)$, where 8^2 is the variance. Express $P(X < 42)$ using the standard normal CDF Φ .
8. A fair die is rolled 5 times. Let X be the number of sixes. Find $P(X = 2)$.
9. A fair coin is flipped until the first head. Find the probability that the first head occurs on the third flip.
10. Explain whether the situation in Exercise 8 is binomial or geometric.
11. A weighted spinner has outcomes a, b, c with probabilities 0.2, 0.5, 0.3. Find $P(\{a, c\})$ and explain why $2/3$ is not correct.
12. In a constructed survey of 40 students, 24 study music, 18 study art, and 10 study both. Translate “exactly one of music and art” into event notation and find its probability.
13. A table has 12 athletes who like chess, 8 athletes who do not like chess, 6 non-athletes who like chess, and 14 non-athletes who do not like chess. Let A be “athlete” and H be “likes chess.” Find $P(H | A)$ and $P(A | H)$.
14. A population has 30% users of a service. A survey response is positive for 80% of users and 20% of nonusers. Find the probability of a positive response and then $P(\text{user} | \text{positive})$.
15. A bag contains 3 red and 2 blue marbles. Compare $P(RR)$ for two draws with replacement and without replacement.
16. A code has three positions and each position can use one of the digits 0–9. How many codes are possible if repetition is allowed? How many if repetition is prohibited?
17. For two fair coin flips, let X be the number of heads. Give the support of X and its PMF.
18. If $X \sim U(0, 10)$, find $P(2 < X < 7)$.
19. For $X \sim \text{Bin}(5, 1/6)$, find $P(X \geq 1)$.
20. For a geometric random variable with $p = 1/6$, find $P(X > 5)$.
21. Decide whether drawing 4 cards without replacement and counting hearts is exactly binomial. Explain using the binomial checklist.
22. Choose the most appropriate method for each task and justify briefly:
 - (a) probability of a password when order matters and repetition is allowed;
 - (b) probability of a positive test after base rates and conditional test rates are given;
 - (c) probability that a continuous waiting time lies between two numbers;
 - (d) probability that the first success occurs on trial 6.

Solutions

1. $\hat{P}(\text{heads}) = 11/20 = 0.55 = 55\%$.
2. $P(\text{not king}) = 1 - 4/52 = 48/52 = 12/13$.
3. Use the union rule:

$$P(C \cup B) = \frac{18 + 12 - 7}{30} = \frac{23}{30} \approx 0.7667.$$

4. The draws are dependent:

$$P(\text{two red}) = \frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15} \approx 0.1333.$$

5. This is a permutation because medal order matters:

$${}_{12}P_3 = 12 \cdot 11 \cdot 10 = 1320.$$

6. This is a combination because committee order does not matter:

$$\binom{9}{4} = 126.$$

7. Standardize using standard deviation $\sigma = 8$:

$$z = \frac{42 - 50}{8} = -1, \quad P(X < 42) = \Phi(-1) \approx 0.1587.$$

8. Here $X \sim \text{Bin}(5, 1/6)$, so

$$P(X = 2) = \binom{5}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 = \frac{625}{3888} \approx 0.1608.$$

9. The first two flips must be tails and the third must be heads:

$$P(X = 3) = \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) = \frac{1}{8} = 0.125.$$

10. Exercise 8 is binomial because there is a fixed number of trials, 5, and we count the number of successes. Exercise 9 is geometric because trials continue until the first success.
11. Add the weights of the favorable elementary outcomes:

$$P(\{a, c\}) = 0.2 + 0.3 = 0.5.$$

The answer $2/3$ incorrectly treats a, b, c as equally likely.

12. Exactly one is $(M \cap A^c) \cup (M^c \cap A)$, where M is music and A is art. The number studying exactly one is

$$(24 - 10) + (18 - 10) = 14 + 8 = 22,$$

so the probability is $22/40 = 11/20 = 0.55$.

13. The total number of athletes is $12 + 8 = 20$, and the total number who like chess is $12 + 6 = 18$. Therefore

$$P(H | A) = \frac{12}{20} = \frac{3}{5} = 0.6, \quad P(A | H) = \frac{12}{18} = \frac{2}{3} \approx 0.6667.$$

The reversed conditionals differ because their denominators differ.

14. Let U be user and $+$ be positive. By total probability,

$$P(+) = (0.30)(0.80) + (0.70)(0.20) = 0.24 + 0.14 = 0.38.$$

Bayes' theorem gives

$$P(U | +) = \frac{(0.30)(0.80)}{0.38} = \frac{0.24}{0.38} = \frac{12}{19} \approx 0.6316.$$

15. With replacement,

$$P(RR) = \frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25} = 0.36.$$

Without replacement,

$$P(RR) = \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{10} = 0.3.$$

The second branch probability changes without replacement.

16. With repetition allowed, there are $10^3 = 1000$ codes. Without repetition, there are

$${}_{10}P_3 = 10 \cdot 9 \cdot 8 = 720$$

codes.

17. The support is $\{0, 1, 2\}$, and

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}.$$

These probabilities sum to 1.

18. For a uniform distribution on $[0, 10]$,

$$P(2 < X < 7) = \frac{7 - 2}{10 - 0} = \frac{1}{2} = 0.5.$$

19. Use the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{5}{6}\right)^5 = \frac{4651}{7776} \approx 0.5981.$$

20. For the first-success convention,

$$P(X > 5) = \left(1 - \frac{1}{6}\right)^5 = \left(\frac{5}{6}\right)^5 = \frac{3125}{7776} \approx 0.4019.$$

21. It is not exactly binomial. The number of draws is fixed and each draw can be classified as heart or not heart, but without replacement the draws are dependent and the probability of a heart changes after each draw.

22. The appropriate methods are:

- (a) ordered counting with repetition, using n^r ;
- (b) total probability and Bayes' theorem, because base rates and conditional rates are given;
- (c) a continuous density or CDF calculation, because the variable is continuous and the event is an interval;
- (d) a geometric model, because the question asks for the first success on a specified trial.

Table 15: Core formulas from the ten topics.

Topic	Formula and conditions
Foundations	$P(A) \geq 0$, $P(S) = 1$, $P(\emptyset) = 0$; if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$
Consequences	$P(A^c) = 1 - P(A)$; if $A \subseteq B$, then $P(A) \leq P(B)$
Experimental probability	$\hat{P}(A) = n(A)/N$, after $N > 0$ observed trials
Equally likely outcomes	$P(A) = A / S $, finite equally likely sample space
Weighted finite space	$P(A) = \sum_{\omega \in A} P(\{\omega\})$, elementary probabilities nonnegative and summing to 1
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
De Morgan's laws	$(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
Conditional probability	$P(B A) = P(A \cap B) / P(A)$, $P(A) > 0$
Multiplication law	$P(A \cap B) = P(A) P(B A) = P(B) P(A B)$ when defined
Chain rule	$P(A_1 \cap \dots \cap A_m) = P(A_1) P(A_2 A_1) \dots P(A_m A_1 \cap \dots \cap A_{m-1})$
Independence	$P(A \cap B) = P(A) P(B)$; if denominators are positive, $P(B A) = P(B)$ and $P(A B) = P(A)$
Total probability	If $B_1, \dots, B_m$ partition S , then $P(A) = \sum_i P(B_i) P(A B_i)$
Bayes' theorem	$P(B_j A) = \frac{P(B_j) P(A B_j)}{\sum_i P(B_i) P(A B_i)}$, $P(A) > 0$
Counting principle	$m_1 m_2 \dots m_r$ choices over r stages
Ordered with repetition	n^r ordered selections when each position has n choices
Permutations	${}_n P_r = n! / (n - r)!$, order matters, no repetition
Combinations	$\binom{n}{r} = n! / [r!(n - r)!]$, order does not matter; ${}_n P_r = r! \binom{n}{r}$
PMF and CDF	$p_X(x) = P(X = x)$, $p_X(x) \geq 0$, $\sum_x p_X(x) = 1$; $F(x) = P(X \leq x)$; $P(X > x) = 1 - F(x)$
Expected value and spread	$E(X) = \sum_x x P(X = x)$ for discrete X ; $\text{Var}(X) = E[(X - E(X))^2]$; $\text{SD}(X) = \sqrt{\text{Var}(X)}$
Continuous density	$f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$, $P(a \leq X \leq b) = \int_a^b f(x) dx$, and $P(X = x) = 0$
Uniform	$X \sim U(a, b)$ has $f(x) = 1/(b - a)$ for $a \leq x \leq b$
Normal	$\Phi(z) = P(Z \leq z)$; if $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$
Normal CDF identities	$P(Z > z) = 1 - \Phi(z)$, $P(a < Z < b) = \Phi(b) - \Phi(a)$, $\Phi(-z) = 1 - \Phi(z)$
Binomial	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, \dots, n$, for fixed independent Bernoulli trials with constant p
Binomial mean/spread	$E(X) = np$, $\text{Var}(X) = np(1 - p)$, $\text{SD}(X) = \sqrt{np(1 - p)}$; $P(X \leq r) = \sum_{k=0}^r P(X = k)$; $P(X \geq 1) = 1 - P(X = 0)$
Geometric	$P(X = k) = (1 - p)^{k-1} p$, $k = 1, 2, \dots$, for the trial number of the first success
Geometric tail/mean/spread	$P(X > k) = (1 - p)^k$, $P(X \leq k) = 1 - (1 - p)^k$, $E(X) = 1/p$, $\text{Var}(X) = (1 - p)/p^2$